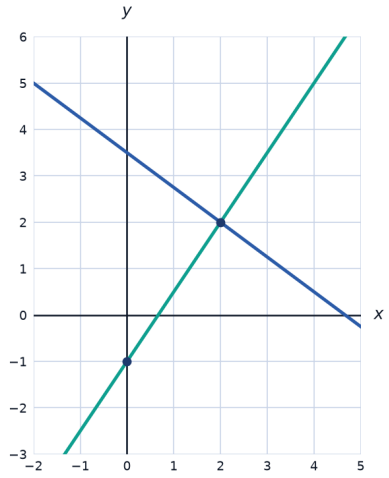
Exponential growth and decay

Core Concepts: Exponential growth and decay

1 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(2)$.



a. 15

b. 21

c. 12

d. 19

2 Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

3 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(3)$.

a. 21

b. 27

c. 18

d. 25

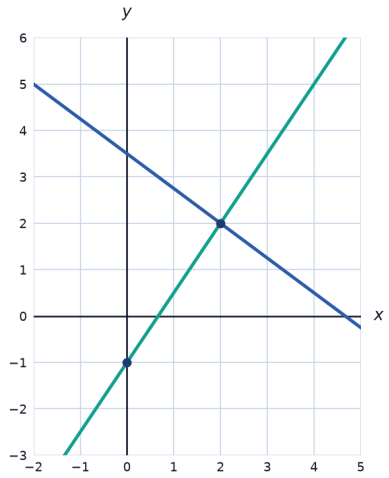
5 Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$



6 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(4)$.

a. 27

b. 33

c. 24

d. 31

8 Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

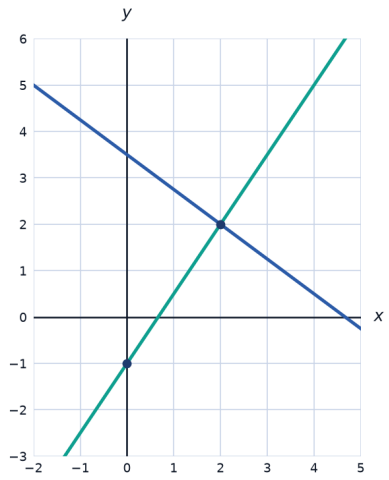
9 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

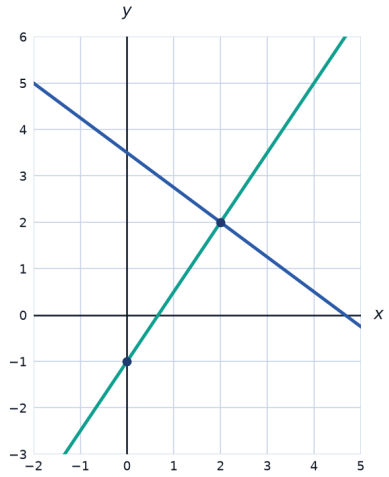
d. 22.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(5)$.



a. 33

b. 39

c. 30

d. 37

11 Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

12 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

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